

YEAR 7 · UNIT 2 · HOMEWORK 7

What Kind, and How Many

Mathematics 2.3–2.4 – collecting like terms, expanding brackets

Science 2.5–2.6 – atoms, elements, compounds and formulae

Section E – the two rules both subjects taught you this week

Name: _____

Class: _____

Date given: _____

Date due: **Monday 21 September 2026****How to do this homework**Five sections. Maths first, then science, then **Section E**, where the two meet.

- In maths, $3x + 5$ is a **finished answer**. You are allowed to stop.
- In science, write **full English sentences**. “Compound” is not a sentence; “Salt is a compound of sodium and chlorine” is.
- **No calculator.**

Section A · Mathematics 2.3 – Collecting Like Terms

A1 How many terms?

Count the terms in each expression. The first one is done for you.

What a term is

A **term** is one part of an expression. The **plus and minus signs separate the terms**. A number on its own is a term too. So $4a + 3b$ has **two** terms: $4a$ and $3b$.

Expression	Terms	Expression	Terms
$4a + 3b$	2	$8m - 3m + n$	
$5x$		6	
$2p + 7q + 4$		$y + 2z - 5 + 3y$	

A2 Like or not like?

Tick one box in each row.

Word help – one word with two jobs

like every day, to enjoy something. In maths, **the same kind**: *milk is like water – both are liquids.*

like terms terms with **the same letter**, so they can be added together

	Like	Not like		Like	Not like
$3a$ and $5a$			$2ab$ and $5ba$		
$4x$ and $4y$			m and m^2		
7 and 2			$5c$ and $-2c$		

A3 Simplify

Collect the like terms. Two of these are already finished – leave them alone.

x means 1x

A letter on its own already has a **1** in front of it. Nobody writes the 1, but it is there.

$$8s - s = 8s - 1s = 7s \quad \text{not 8.}$$

(a) $2b + 3b =$ _____

(f) $4x + x + 2x =$ _____

(b) $9k - 4k =$ _____

(g) $3a + 7 =$ _____

(c) $6m + 3n + 2m =$ _____

(h) $6y - y + 2y =$ _____

(d) $5p + 2q + p + 4q =$ _____

(i) $10g - 6g - g =$ _____

(e) $8s - s =$ _____

(j) $5c + 3d =$ _____

A4 Every sign travels with its own term

Take the sign with you

The + or – in front of a term **belongs to that term**. When you move the term, the sign goes with it.

$$7x + 5y - 3x + y = 7x - 3x + 5y + y = 4x + 6y$$

Simplify. Write the middle line, with the like terms next to each other.

(a) $9x + 4y - 5x + 2y$ _____

(b) $8a - 3b + 2a - b$ _____

(c) $6 + 5t - 1 - 2t$ _____

(d) $7p - 2q - 3p + 6q + 4$

A5 Now you write the question**Read all four conditions before you start**

Here is an expression:

$5a + 3b$

Write a **word problem** that this expression is the answer to. Your question must:

1. be about something **real** — a shop, a classroom, a lab bench;
2. use **two different kinds** of thing, one for a and one for b ;
3. say clearly what a and b **stand for**;
4. be written in **full sentences** and **end in a question**.

Section B · Mathematics 2.4 — Expanding Brackets**B1 Fill the grid**

One box, one multiplication. Fill in the empty boxes, then write the answer.

Word help — the words that carry the whole rule

- expand · multiply out** the same instruction in different words. Both appear in the exercise
- each / every** not a maths word, but it carries the whole rule: **every** term inside, not just the first

Brackets mean multiply $4(x + 3)$ means $4 \times (x + 3)$. The times sign is hidden, exactly as it is in $4n$.

$\times$	x	$+ 3$
4	$4x$	$+ 12$

$4(x + 3) = 4x + 12$

Not $4x + 3$. The 3 is inside the brackets, so it gets multiplied too.

×	m	$+ 4$
3		

(a) $3(m + 4) =$ _____

×	y	$+ 2$
5		

(b) $5(y + 2) =$ _____

×	$2n$	$+ 3$
4		

(c) $4(2n + 3) =$ _____

B2 Expand

No grid this time. Draw one if it helps.

Two things that go inside with the term**The minus sign.** $3(x - 2) = 3x - 6$ and $7(1 - w) = 7 - 7w$.**A number already in front of the letter.** Multiply the numbers, keep the letter: $5 \times 2p = 10p$, **not** $52p$.

(a) $3(a + 2) =$ _____

(e) $8(7 + z) =$ _____

(i) $4(3 - t) =$ _____

(b) $5(b + 3) =$ _____

(f) $6(k - 3) =$ _____

(j) $3(2q + 5) =$ _____

(c) $9(y + 3) =$ _____

(g) $2(y - 4) =$ _____

(k) $4(5u - 1) =$ _____

(d) $4(2 + f) =$ _____

(h) $5(m - 6) =$ _____

(l) $6(1 + 2v) =$ _____

(m) This one has three terms inside. $8(6 + 4w - 3g) =$ _____

B3 Knowing when to stop

Tick one box in each row. If it can be collected, write the answer.

Expanding does not switch off Section A

After you expand, look for like terms. Often there are none.

 $4(3 - c) = 12 - 4c$, and 12 and $4c$ are **not like terms**. So $12 - 4c$ is **finished**.

Expression	Finished	Can be collected	If so, write it
$12 - 4c$			
$5x + 3x$			
$3a + 6$			
$8 + 2m + 5$			

B4 Mr Bowen's homework, again

One of these four is right. Find it, and correct the other three.

(a) $3(a + 4) = 3a + 4$ _____

(c) $2(5 - d) = 10 - 2d$ _____

(b) $4(2b - 5) = 8b - 9$ _____

(d) $6(m + 2) = 6m + 8$ _____

B5 Expand, then collect

Two jobs, in this order. Expand the brackets first, then look for like terms.

(a) $3(2y + 1) + 5y + 4$

(b) $2(a + 6) + 3(a + 1)$

B6 The fish tanks

Mr Bowen keeps **7** fish tanks in his kitchen. Every tank holds f fish and **2** rubber ducks.

(a) Write an expression with brackets for the number of things in the tanks altogether.

(b) Expand it.

(c) Rubber ducks are not alive. Write an expression for the number of **living** things in the tanks.

Section C · Science 2.5 — Atoms, Elements and the Periodic Table

C1 Use the Periodic Table in your book

Groups are numbered 1 to 8.

Element	Symbol	Period	Group	Metal or non-metal
magnesium				
chlorine				

C2 Match the word to its meaningWrite the correct **letter** in each box.

Word		Meaning
1. Atom		A A column in the Periodic Table.
2. Element		B A short way to write the name of an element.
3. The Periodic Table		C Elements such as iron, copper and aluminium.
4. Period		D A tiny piece of matter. Everything is made of them.
5. Group		E A very, very small tube made of carbon atoms.
6. Symbol		F A substance made of only one kind of atom.
7. Nanotube		G A row in the Periodic Table.
8. Metals		H A way of arranging all the elements.

C3 The capital-letter rule

(a) Four students wrote a symbol wrongly. Write it correctly, and name the element.

One capital, then small letters

An element symbol starts with a **CAPITAL** letter. If there is a second letter it is **small**.
Mg is magnesium — **one** element. MG and mg are not symbols at all.

Written	Correct	Element	Written	Correct	Element
NA			mG		
cl			CA		

(b) Mr Bowen means cobalt, and writes CO. Cobalt is Co. Explain what he has actually written down.

C4 Answer in full sentences

Diamond is hard and clear. Graphite is soft and grey. Both are made of **only** carbon atoms. Is diamond an element? Explain your answer.

Sentence starters — you may copy these

“An element is a substance that...” “...because every atom in it is...”

Section D · Science 2.6 – Compounds and Formulae

D1 Element or compound?

Tick one box in each row.

Count the kinds, not the atoms

An **element** is made of **one kind** of atom. A **compound** is made of **two or more kinds** of atom **bonded** together. O_2 has two atoms and they are **both oxygen** – one kind, so it is an **element**.

	Element	Compound		Element	Compound
Na			H_2		
NaCl			$CaCO_3$		
O_2			Al		
CO_2			H_2O		

D2 Naming a compound

Name the compound made from each set of elements.

Word help – the endings and the counting words

-ide two elements joined. The metal comes first; the non-metal ends in **-ide**
-ate two elements and **oxygen** joined. The ending is often **-ate**
mono / di one / two. CO is carbon monoxide; CO_2 is carbon dioxide

(a) sodium and chlorine

(c) lithium and sulfur

(b) magnesium and oxygen

(d) calcium, carbon and oxygen

(e) Mr Bowen writes *sulfur calcide*. Two things are wrong. Write the correct name and say what the two mistakes were.

D3 A compound is nothing like the elements in it

Sodium is a soft metal that bursts into flame in water. Chlorine is a poisonous yellow-green gas. Bonded together they make sodium chloride, which you ate with your dinner last night.

A student says: *"Salt must be dangerous, because chlorine is poisonous."* Explain, in full sentences, why this is wrong.

D4 Draw the particles

Draw the particles of each substance in the box. Use a different colour or pattern for each kind of atom, then tick element or compound.

oxygen O ₂	water H ₂ O	carbon dioxide CO ₂	methane CH ₄
element ☐ compound ☐	element ☐ compound ☐	element ☐ compound ☐	element ☐ compound ☐

Section E • Where the two subjects meet

In maths a letter stands for a number. In science a letter stands for an element. You read them with the same two rules.

The two rules

- A letter says what kind. A number says how many.** $3a \cdot O_3$
- A number written in front multiplies everything after it.** $4(x + 3) \cdot 3CO_2$

E1 Mg or mg?

Complete the table. The first row is done for you.

The capital is doing a job

In **maths** every letter is small, and two letters written together means **multiply**: mg means $m \times g$.

In **science** a symbol begins with a **capital**: Mg is magnesium, one element. Two capitals means **two** elements: CO is carbon and oxygen.

Written	Maths or science?	What it means
mg	maths	m multiplied by g
Mg		
CO		
Co		
xy		
$NaOH$		

E2 The 1 that nobody writes – in both subjects

Write the number that is **not printed**. The first one is done for you.

Same habit, two subjects

In **maths**, s means $1s$ – that is why $8s - s = 7s$. In **science**, H_2O has nothing after the O , so there is **one** oxygen atom. Both subjects leave the 1 out and expect you to know it is there.

Written	The question	The number
$8s - s$	how many s is the second term?	1
$w + w$	how many w is each term?	
H_2O	how many oxygen atoms?	
$NaCl$	how many chlorine atoms?	

E3 Counting atoms is expanding brackets

Complete the table. The first row is done for you.

The same grid, on a formula

$3CO_2$ means **three lots of** CO_2 . One CO_2 is one carbon atom and two oxygen atoms. So put it in the grid:

×	C	+ 2 O
3	3 C	+ 6 O

$$3(C + 2O) = 3C + 6O$$

Three CO₂ is **3 carbon atoms and 6 oxygen atoms**.

Formula	Atoms of each element	Atoms in total
3CO ₂	3 carbon, 6 oxygen	9
2H ₂ O		
4NaCl		
2CaCO ₃		

(b) Now write 2H₂O the way a maths teacher would. Fill the boxes.

$$2(2H + O) = \square H + \square O$$

E4 The same mistake, in two subjects

Mr Bowen made two mistakes on the same day. In **maths** he wrote

$$x + x = x^2$$

In **science** he said: "O₂ has two atoms, so it is a compound."

(a) Write the correct answer to $x + x$.

(b) Is O₂ an element or a compound?

(c) These two mistakes are the **same** mistake. Explain what it is, in full sentences.

E5 And he made it again

A week later Mr Bowen wrote these two lines, one in each book.

Maths: $4(x + 3) = 4x + 3$

Science: 6CO₂ is 6 carbon atoms and 2 oxygen atoms.

(a) Write the maths line correctly.

(b) Write the science line correctly.

(c) What did he forget to do, both times? Answer in one sentence.

Answer Key – Teacher Copy

A1 How many terms?

$4a + 3b$: 2 $5x$: 1 $2p + 7q + 4$: 3 $8m - 3m + n$: 3 6: 1 $y + 2z - 5 + 3y$: 4

The two rows that separate the class are 6 and $2p + 7q + 4$. A number on its own is a term, and students who have decided a term must contain a letter write 0 for the first and 2 for the second. Both errors come from one belief, so fix it once.

A2 Like or not like?

Left column: $3a$, $5a$ like · $4x$, $4y$ not · 7, 2 like.

Right column: $2ab$, $5ba$ like · m , m^2 not · $5c$, $-2c$ like.

$2ab$ and $5ba$ is the row worth the time: multiplication can be written in either order, so they are the same kind. If anyone doubts it, put $a = 2$ and $b = 3$ in both.

$5c$ and $-2c$ is like. The sign belongs to the term but does not change what kind of term it is – which is exactly the fact A4 needs.

m and m^2 is the only row that cannot be settled with an everyday picture. m^2 means $m \times m$, and that is a different thing from m .

A3 Simplify

(a) $5b$ (b) $5k$ (c) $8m + 3n$ (d) $6p + 6q$ (e) $7s$

(f) $7x$ (g) $3a + 7$ **already finished** (h) $7y$ (i) $3g$ (j) $5c + 3d$ **already finished**

(e), (f), (h) and (i) all contain a lone letter, and skipping it goes wrong in two directions, which is worth knowing before you mark. Where the lone letter is **added**, the answer comes out one short: (f) gives $6x$. Where it is **subtracted**, the answer comes out one too many: (h) gives $8y$ and (i) gives $4g$. (e) is the extreme case – $8s - s$ becomes 8, with the letter gone altogether. Ask for the 1 in pencil until it stops being needed.

(g) and (j) are the two that must be left alone, and the instruction line says there are two. A student who simplifies (g) to $10a$ has not read it; a student who leaves eight alone has read it too carefully.

(h) is the hardest item: the invisible 1 is in the middle of the line *and* carries a minus sign.

A4 Every sign travels with its own term

(a) $9x - 5x + 4y + 2y = 4x + 6y$ (b) $8a + 2a - 3b - b = 10a - 4b$

(c) $5t - 2t + 6 - 1 = 3t + 5$ (d) $7p - 3p - 2q + 6q + 4 = 4p + 3q + 4$

Mark the middle line, not the answer. Every error here is invisible in a final number and obvious in the working, which is the whole argument for demanding it.

(b) is the one that goes wrong: the $-b$ is a lone letter *and* carries a minus, so it collects two errors at once and comes out $10a - 2b$ or $10a - 4$.

(d) ends with a number that belongs to nothing, and a class that has just learned to collect will try to collect it.

A5 Now you write the question

No single right answer. Check the four conditions **in this order**:

1. **Are there two different kinds of thing?** This is the one that fails. A question about 5 red pens and 3 blue pens is still one kind of thing being counted, and it gives $8p$, not $5a + 3b$.
2. **Is it said what a and b stand for?**
3. **Is each letter a number, not an object?** “ a is a mango” is the error; “ a is the number of mangoes in one bag” is right.

4. Full sentences, ending in a question?

One that works: "A bag holds a apples. A box holds b bananas. Mr Bowen buys 5 bags and 3 boxes. Write an expression for the number of pieces of fruit he has."

A question ending "what is $5a + 3b$?" has not done the task — it is the expression read out in words. Name that out loud rather than marking it wrong.

Read four or five aloud. This is the best item in Section A for finding out who is reading.

B1 Fill the grid

(a) $3m + 12$ (b) $5y + 10$ (c) $8n + 12$

(c) is the only one where the left-hand box is not a lone letter, and it is the one to check: $4 \times 2n$ is $8n$, not $6n$ and not $8n^2$. That is B2's second rule arriving before it is stated, on purpose — the grid makes it obvious and the rule can be named afterwards.

Strong students will start writing the answer straight out. That is fine; ask them to draw the grid for (c) anyway.

B2 Expand

(a) $3a + 6$ (b) $5b + 15$ (c) $9y + 27$ (d) $8 + 4f$ (e) $56 + 8z$ (f) $6k - 18$
 (g) $2y - 8$ (h) $5m - 30$ (i) $12 - 4t$ (j) $6q + 15$ (k) $20u - 4$ (l) $6 + 12v$
 (m) $48 + 32w - 24g$

(d) and (e) put the number first inside the brackets, and a student who has learned "letter then number" writes $4f + 8$. That is the same expression, so accept it and say why out loud.

(i) is the one to watch: the letter is second, the answer starts with a number and *looks* unfinished — and B3 is about to say that it is not.

(m) has three terms and the third gets left behind. Counting the terms inside before starting is A1 doing real work.

B3 Knowing when to stop

$12 - 4c$ finished $5x + 3x \rightarrow 8x$ $3a + 6$ finished $8 + 2m + 5 \rightarrow 2m + 13$

$8 + 2m + 5$ is the useful row. The two numbers are *not* next to each other, and a student reading left to right will not see them as a pair. It is A4's skill in B's clothing.

This question is why Section A comes before Section B. A student who can expand perfectly and cannot do this has learned two lessons and not joined them.

B4 Mr Bowen's homework, again

(a) **wrong** — $3a + 12$. He multiplied the a and not the 4.

(b) **wrong** — $8b - 20$. He did $4 + 5$ instead of 4×5 .

(c) **right**.

(d) **wrong** — $6m + 12$. He did $6 + 2$ instead of 6×2 .

(b) and (d) are the same error and are printed apart on purpose: a student who corrects one and not the other is guessing.

(a) is the headline error of 2.4, and it reappears in E5 wearing a chemical formula. If you mark one question in Section B, mark this one, and keep the answers to show alongside E5.

B5 Expand, then collect

(a) $6y + 3 + 5y + 4 = 11y + 7$ (b) $2a + 12 + 3a + 3 = 5a + 15$

The order is the lesson: expand, *then* collect. A student who tries to collect first meets $2y + 1$ inside a bracket and stalls.

(b) has two brackets and is the only place in the packet where one expansion has to be held while another is done. Insist on both being written out before anything is collected.

B6 The fish tanks(a) $7(f + 2)$ (b) $7f + 14$ (c) $7f$

(c) is the best item in Section B. The answer is $7f$, with no number added and no bracket — fourteen rubber ducks simply are not alive. Expect blanks, and expect $7f + 14$. Anyone who writes $7f$ has read every word.

An expression that is a single term with no operation in it does not look like an answer to this class. Say that it is one.

C1 Use the Periodic Table in your bookmagnesium Mg, period 3, group 2, **metal**chlorine Cl, period 3, group 7, **non-metal**

The two are in the same period and different groups on purpose: one row across, two columns apart, so a student who has period and group the wrong way round gives 2 and 3, and 7 and 3, and is wrong in a way you can read at a glance.

This is the only question on the packet that cannot be answered without opening the book. A blank here may mean the book stayed in the bag rather than that the science is missing — ask before marking it.

Group numbers follow the book's table, which numbers them 1 to 8. A student using a wall chart numbered 1 to 18 will say 17 for chlorine. That is a different table, not a wrong answer; ask which one they used.

C2 Match the word to its meaning

1-D 2-F 3-H 4-G 5-A 6-B 7-E 8-C

Period (4-G) and *group* (5-A) are the pair that swap, and not because of the science: both words have an everyday English meaning — a lesson, and people together — and neither helps. Make the class say **period is a row, group is a column** out loud.

A student with 4 and 5 reversed and everything else right has a vocabulary problem, not a chemistry problem.

C3 The capital-letter rule(a) $NA \rightarrow Na$, sodium $cl \rightarrow Cl$, chlorine $mG \rightarrow Mg$, magnesium $CA \rightarrow Ca$, calcium

(b) CO is **two** symbols, C and O, so he has written carbon and oxygen bonded together — a compound, not an element. Cobalt is Co, with a small second letter.

(b) is the question. The capital letter is not neatness, and this is the one place a student can see that it changes the meaning completely: CO and Co are different substances.

It is also E1 in advance, so keep the answers.

(c) Magnesium is in **period 3, group 2**, and it is a **metal**.

(c) is the only question on the packet that cannot be answered without opening the book, so a blank here may mean the book stayed in the bag rather than that the science is missing. Ask before marking.

Group numbers follow the book's table, which numbers them 1 to 8. A student using a wall chart numbered 1 to 18 will still say 2 for magnesium — but for carbon or chlorine the two systems disagree, so if this question is ever widened, say which table to use.

C4 Answer in full sentences

Yes. Diamond is made of only one kind of atom — carbon — so it is an element. Diamond and graphite look different because the carbon atoms are **joined in different arrangements**, not because they are different elements.

Discuss this one rather than marking it. The everyday evidence — one hard and clear, one soft and grey — points the wrong way, and a student who says “no, they look different” has reasoned from what they

can see. Say that is good science, then give them the extra fact.

Watch for “an element has one atom”. One *kind* of atom; a gold ring has an enormous number of them.

D1 Element or compound?

Elements: Na, O₂, H₂, Al. Compounds: NaCl, CO₂, CaCO₃, H₂O.

O₂ and H₂ are the whole question, and most of the class will tick compound for both. Two atoms is not two kinds.

Do not mark these two rows — put them on the board and take a hand vote before saying anything. It is the same misconception as E4, and it is worth meeting twice in one evening.

D2 Naming a compound

(a) sodium chloride (b) magnesium oxide (c) lithium sulfide (d) calcium carbonate

(e) **Calcium sulfide**. Two mistakes: the **metal must come first** (calcium, not sulfur), and the **non-metal takes the -ide ending** (sulfide, not calcide).

(d) is the only *-ate*, and the trigger is the oxygen. A student who writes calcium carbide knows the ending rule and did not notice the oxygen.

(e) can be got right for the wrong reason. Ask for both mistakes, named separately.

D3 A compound is nothing like the elements in it

When atoms bond, the compound has **new properties** of its own. Sodium chloride is not sodium and it is not chlorine — it is a new substance, and it is safe to eat. What the elements do on their own tells you nothing about what the compound does.

This is the one sentence of 2.6 that has to survive the week. Push for the words *new properties*; “because it is a compound” is circular and is what most of them will write first.

A student who brings up chlorine in swimming pools has made a real connection and is also making the point — pool chlorine is not the chlorine bonded in salt.

D4 Draw the particles

O₂ two circles of one kind, joined — **element**. H₂O one large circle with two small ones joined to it — **compound**.

CO₂ one central circle with two of a different kind, one each side — **compound**. CH₄ one central circle with four of a different kind around it — **compound**.

Mark two things and no more: **are the atoms touching**, and **is each kind drawn differently**. Loose separated circles mean a mixture, which is 2.7 and not what any of these are.

A CH₄ with three hydrogens is not a drawing error — it is the small number not being read, and it is the same failure E3 tests with a big number in front. Do not mark the artwork.

E1 Mg or mg?

Mg — science — magnesium, one element

CO — science — carbon and oxygen, two elements

Co — science — cobalt, one element

xy — maths — x multiplied by y

NaOH — science — sodium, oxygen and hydrogen: **three** elements

Mark CO and Co together. Same two letters, same order, one capital changed, and the answer goes from two elements to one. Nothing in maths behaves like that, which is why the question is here.

NaOH is the hardest row and the one worth the most. Na is *one* symbol made of two letters, so a student who reads four letters as four elements — or as nitrogen, oxygen and hydrogen — has ignored the small *a*. It is the same reading that CaCO₃ needs in E3.

Expect “Mg means *m* times *g*” from at least one student who read the shape and not the case.

E2 The 1 that nobody writes

$w + w - 1$ each $\text{H}_2\text{O} - 1$ oxygen atom $\text{NaCl} - 1$ chlorine atom

This is A3 and the science formulae side by side, and the point is that they are one habit rather than two. Both subjects leave out the 1; neither warns you.

A student who has A3 right and this wrong is not being inconsistent — they have learned a maths rule rather than a reading rule. Show them the two rows together.

E3 Counting atoms is expanding brackets

$2\text{H}_2\text{O} - 4$ hydrogen, 2 oxygen — 6 atoms

$4\text{NaCl} - 4$ sodium, 4 chlorine — 8 atoms

$2\text{CaCO}_3 - 2$ calcium, 2 carbon, 6 oxygen — 10 atoms

(b) $2(2\text{H} + \text{O}) = 4\text{H} + 2\text{O}$

This is the question the packet was built around. The two numbers in $2\text{H}_2\text{O}$ do different jobs: the small 2 counts hydrogen atoms inside one molecule, and the big 2 in front multiplies all of it. That is exactly the difference between the 3 inside $4(x + 3)$ and the 4 outside it.

2CaCO_3 is the hard row, because the big 2 has to reach past two symbols to get to the small 3: $2 \times 3 = 6$ oxygen atoms. A student who writes 3 stopped multiplying halfway along the bracket, which is B4(a) exactly.

(b) is not a trick. Atoms in a formula are counted this way; the brackets are simply not printed.

E4 The same mistake, in two subjects

(a) $x + x = 2x$ (b) **Element.**

(c) Both times he counted **how many** and then answered as though it told him **how many kinds**. Two x is still one kind of term, so they collect into $2x$. Two oxygen atoms are still one kind of atom, so O_2 is still an element. **Two of the same thing is not two different things.**

Discuss this one; do not mark it. It is the only question in the packet that asks the class to see two errors from two subjects as one error, and the wording they reach for will be rough. Accept anything that gets to *same kind*, however it is phrased. "He thought two meant two different" has it.

If the room is silent, put $x + x$ and O_2 on the board side by side and ask only: how many *kinds* are in each? The answer arrives on its own.

E5 And he made it again

(a) $4(x + 3) = 4x + 12$

(b) 6CO_2 is **6 carbon atoms and 12 oxygen atoms.**

(c) He multiplied the **first** thing and stopped. The number in front has to multiply **everything** after it — every term inside the brackets, every atom in the formula.

Put B4(a) and this question next to each other when you hand the packet back. They are the same line written twice, once with a letter in it and once with an element, and a student who corrected B4(a) confidently and left this one alone has learned a maths rule rather than a rule.

(c) is marked on one word: **everything**. *Each* and *every* carry the whole of 2.4, which is why they are in B1's word help.